Grocery App - REST API Documentation

Laravel 12 + Sanctum Backend API Reference for Flutter Mobile & Web

WEB / DESKTOP / IOS BASE URL	ANDROID EMULATOR BASE URL	INTERACTIVE SWAGGER UI
`http://127.0.0.1:8000/api`	`http://10.0.2.2:8000/api`	`http://127.0.0.1:8000/docs/index.ht`

1. Authentication & Standard Request Headers

All API endpoints return JSON. Protected routes use Laravel Sanctum Bearer tokens with a 30-day expiration lifetime. Include the Authorization header on all authenticated requests (and optionally on GET /api/products to populate isFavorite).

STANDARD HTTP HEADERS

```
Accept: application/json
Content-Type: application/json
Authorization: Bearer <your_sanctum_token>
```

2. Complete API Endpoint Reference Table

Method	Endpoint	Auth	Description
POST	/api/auth/register	Public	Register a new user account & receive 30-day token
POST	/api/auth/login	Public	Login with email & password & receive 30-day token
POST	/api/auth/refresh	Bearer	Refresh/reset token (works even if token is expired)
POST	/api/auth/logout	Bearer	Revoke current access token
GET	/api/auth/me	Bearer	Get current authenticated user profile
POST/PUT	/api/auth/profile	Bearer	Update name, email, phone, avatar photo, or password
GET	/api/categories	Public	List all product categories with iconAsset URLs
GET	/api/products	Optional	List all products (?category={id|all} & ?q={search})
GET	/api/products/{id}	Optional	Get single product details by ID
GET	/api/cart	Bearer	Get current user cart items
POST	/api/cart	Bearer	Add product to cart (validates live stockQuantity)
PUT	/api/cart/{productId}	Bearer	Set item quantity (0 removes item; checks live stock)
DELETE	/api/cart/{productId}	Bearer	Remove single product from cart
DELETE	/api/cart	Bearer	Clear all items from cart
GET	/api/favorites	Bearer	List user favorite/wishlisted products
POST	/api/favorites/{id}/toggle	Bearer	Toggle favorite status for a product
GET	/api/orders	Bearer	List user order history (newest first)
POST	/api/orders	Bearer	Place order from cart (decrements stock & clears cart)
GET	/api/orders/{orderId}	Bearer	Get single order details with line items
GET	/api/addresses	Bearer	List saved delivery addresses
POST	/api/addresses	Bearer	Create a new delivery address
PUT	/api/addresses/{id}	Bearer	Update a saved delivery address
DELETE	/api/addresses/{id}	Bearer	Delete a saved delivery address
GET	/api/media/{path}	Public	Stream uploaded images with CORS headers

3. Authentication & Profile Endpoints

POST `/api/auth/register` | `/api/auth/login` — **Auth: Public**

Registers a new user (201 Created) or logs in an existing user (200 OK). For login, only email and password are required.

REQUEST / PARAMETERS

```
{
  "name": "Sok Dara",
  "email": "dara@example.com",
  "password": "secret123"
}
```

RESPONSE JSON

```
{
  "token": "1|abc123xyz...",
  "expiresAt": "2026-12-02T05:50:00.000000Z",
  "user": {
    "id": "1", "name": "Sok Dara",
    "email": "dara@example.com",
    "phone": "012345678", "avatarUrl": null
  }
}
```

POST `/api/auth/refresh` — **Auth: Bearer Token**

Revokes the current (valid or expired) Bearer token and issues a new 30-day token with the same response structure above.

POST/PUT `/api/auth/profile` — **Auth: Bearer Token**

Updates user profile info, uploads/removes profile avatar (multipart/form-data), or updates password.

REQUEST / PARAMETERS

```
{
  "name": "Sok Dara",
  "email": "dara@example.com",
  "phone": "012345678",
  "avatar": "(file: jpg/png/webp, max 4MB)",
  "remove_avatar": false,
  "current_password": "secret123",
  "new_password": "newSecret456"
}
```

RESPONSE JSON

```
{
  "id": "1",
  "name": "Sok Dara",
  "email": "dara@example.com",
  "phone": "012345678",
  "avatarUrl": "http://127.0.0.1:8000/api/media/..."
}
```

4. Catalog: Categories & Products

GET `/api/categories` — **Auth: Public**

Returns all categories ordered by name.

RESPONSE JSON

```
[
  {
    "id": "1",
    "name": "Fresh Fruits",
    "iconAsset": "http://127.0.0.1:8000/api/media/categories/fruits.png"
  }
]
```

GET `/api/products` | `/api/products/{id}` — **Auth: Optional Bearer**

Lists all products or fetches one product. Supports query filters ?category={id|all} and ?q={search}.

REQUEST / PARAMETERS

```
GET /api/products
GET /api/products?category=6
GET /api/products?q=Angkor
GET /api/products/6
```

RESPONSE JSON

```
{
  "id": "6",
  "name": "Angkor Beer",
  "description": "Angkor Beer",
  "images": [
    "http://127.0.0.1:8000/api/media/products/beer.jpg"
  ],
  "price": 12.0, "originalPrice": 18.0,
  "unit": "330ml", "categoryId": "6",
  "stockQuantity": 15, "isFavorite": false
}
```

5. Cart & Live Stock Validation

POST `/api/cart | PUT /api/cart/{productId}` — **Auth: Bearer Token**

Adds to cart or sets exact item quantity (0 removes item). Returns HTTP 422 if product is out of stock or quantity > stockQuantity.

REQUEST / PARAMETERS

```
// POST /api/cart
{ "product_id": 6, "quantity": 2 }

// PUT /api/cart/6
{ "quantity": 3 }

// 422 Out-of-Stock Error:
{ "message": "Only 15 left in stock...",
  "stockQuantity": 15 }
```

RESPONSE JSON

```
[
  {
    "id": "12",
    "product": {
      "id": "6", "name": "Angkor Beer",
      "images": ["http://127.0.0.1:8000/api/media/..."],
      "price": 12.0, "stockQuantity": 15
    },
    "quantity": 2
  }
]
```

6. Favorites / Wishlist

POST `/api/favorites & /api/favorites/{id}/toggle` — **Auth: Bearer Token**

GET /api/favorites returns an array of favorited Product objects. POST toggle flips favorite state.

REQUEST / PARAMETERS

```
GET  /api/favorites
POST /api/favorites/6/toggle
```

RESPONSE JSON

```
{ "isFavorite": true }
```

7. Orders

POST `/api/orders & /api/orders/{id}` — **Auth: Bearer Token**

Places an order from the cart in an atomic transaction: validates & decrements stock_quantity, adds $5.00 delivery fee, clears cart.

REQUEST / PARAMETERS

```
{
  "address_id": 1,
  "address": "#123 Street 271, Phnom Penh"
}
```

RESPONSE JSON

```
{
  "id": "101",
  "subtotal": 24.0, "deliveryFee": 5.0, "total": 29.0,
  "status": "pending",
  "addressId": "1",
  "address": "#123 Street 271, Phnom Penh",
  "createdAt": "2026-11-02T05:50:00.000000Z",
  "items": [
    { "productId": "6", "name": "Angkor Beer",
      "price": 12.0, "quantity": 2 }
  ]
}
```

8. Saved Delivery Addresses

POST `/api/addresses & /api/addresses/{id}` — **Auth: Bearer Token**

Manage user delivery addresses (GET, POST, PUT, DELETE). Setting is_default=true automatically unsets previous defaults.

REQUEST / PARAMETERS

```
{
  "label": "Home",
  "recipient_name": "Sok Dara",
  "phone": "012345678",
  "address": "#123 Street 271",
  "city": "Phnom Penh",
  "is_default": true
}
```

RESPONSE JSON

```
{
  "id": "1",
  "label": "Home",
  "recipientName": "Sok Dara",
  "phone": "012345678",
  "address": "#123 Street 271",
  "city": "Phnom Penh",
  "isDefault": true
}
```